

Bayes Estimation

Outline

- 1) Bayes risk, Bayes estimator
- 2) Examples
- 3) Conjugate priors

Frequentist Motivation

Model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ for data X

Loss $L(\theta, d)$, Risk $R(\theta; \delta) = \mathbb{E}_\theta[L(\theta, \delta(x))]$

The Bayes risk is the average-case risk, integrated wrt some measure Λ , called prior

For now, assume $\Lambda(\Theta) = 1$ (prob. meas.)

Later we will allow to be improper ($\Lambda(\Theta) = \infty$)

Note: • Λ and $c\Lambda$ for $c > 0$ functionally equiv.

• avg risk makes sense even if we don't "believe" $\theta \sim \Lambda$

$$\begin{aligned} r_\Lambda(\delta) &= \int_{\Theta} R(\theta, \delta) d\Lambda(\theta) \\ &= \mathbb{E}_{\theta \sim \Lambda} [\underbrace{R(\theta, \delta)}_{\mathbb{E}[L(\theta, \delta(x)) | \theta]}] \text{ where } \theta \sim \Lambda \\ &= \mathbb{E} [L(\theta, \delta(x))] \text{ where } \theta \sim \Lambda, \underbrace{x | \theta \sim P_\theta} \end{aligned}$$

$\mathbb{E}$ now means wrt joint distr. of (θ, x)

An estimator δ minimizing $r_\Lambda(\cdot)$ is called

Bayes (a Bayes estimator). Dep. on $\mathcal{P}, \Lambda, L$

$$r_\Lambda(\delta) = \mathbb{E} \left[\underbrace{\mathbb{E}[L(\theta, \delta(x)) | x]}_{\text{minimize one } x \text{ at a time}} \right]$$

Note: we choose this after seeing x

Prior, Posterior

Usual interp. of Δ is "prior belief about θ before seeing the data"

Conditional dist. $\Delta(\theta | x)$ called posterior dist.
"belief after seeing the data"

Epistemic uncertainty:

"I think there is a 50% chance that..."
More on this next time

Densities: prior $\lambda(\theta)$, likelihood $p_\theta(x)$
or $p(x|\theta)$

Joint density $\lambda(\theta) p_\theta(x)$

Marginal density $q(x) = \int_{\Theta} \lambda(\theta) p_\theta(x) d\theta$

Posterior density $\lambda(\theta|x) = \frac{\lambda(\theta) p_\theta(x)}{q(x)}$

Bayes estimator depends on posterior:

$$\begin{aligned} \delta_\Delta(x) &= \operatorname{argmin}_d \mathbb{E}[L(\theta, d) | x] \\ &= \operatorname{argmin}_d \int_{\Theta} L(\theta, d) \lambda(\theta|x) d\theta \end{aligned}$$

Solve for Bayes estimator "one x at a time"

Bayes Estimator

Model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, (proper) prior Δ

Assume

- $L(\theta, d) \geq 0$
- $r_\Delta(\delta_0) < \infty$ for some $\delta_0(x)$

Then $\delta_\Delta(x)$ is Bayes, with $r_\Delta(\delta_\Delta) < \infty$

iff $\delta_\Delta(x) \in \operatorname{argmin}_d \mathbb{E}[L(\theta, d) | X=x]$ a.e. x

Proof

$\mathbb{P}(\delta_\Delta(x) \neq \operatorname{argmin}) = 0$
wrt marginal

($\Leftarrow$) δ any other estimator

$$\mathbb{E}[L(\theta, \delta(x)) | X] \stackrel{\text{a.s.}}{\geq} \mathbb{E}[L(\theta, \delta_\Delta(x)) | X]$$

Marginalize $\leadsto r_\Delta(\delta) \geq r_\Delta(\delta_\Delta) < \infty$ (take $\delta = \delta_0$)

($\Rightarrow$) Assume $r_\Delta(\delta_\Delta) < \infty$ (o.w. δ_0 better)

Let $E_x(d) = \mathbb{E}[L(\theta, d) | X]$, $E_x^* = \inf_d E_x(d)$

For $\varepsilon > 0$, $A_\varepsilon = \{x : E_x^* < E_x(\delta_\Delta(x)) - \varepsilon\}$

Must be some $\varepsilon > 0$ with $\mathbb{P}(A_\varepsilon) > 0$

On A_ε , choose $\delta^*(x)$ so $E_x(\delta^*(x)) \leq E_x(\delta_\Delta(x)) - \varepsilon$

on A_ε^c , $\delta^*(x) = \delta_\Delta(x)$

$$\begin{aligned} r_\Delta(\delta_\Delta) - r_\Delta(\delta^*) &= \mathbb{E}[E_x(\delta_\Delta(x)) - E_x(\delta^*(x))] \\ &\geq \varepsilon \mathbb{P}(A_\varepsilon) \end{aligned}$$

Bayes Estimator

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($\Rightarrow$) Assume $r_\Delta(\delta_\Delta) < \infty$ (o.w. δ_0 better)

Define $E_x(d) = \mathbb{E}[L(\theta, d) | X]$

$$A_\varepsilon = \{x : E_x(\delta_\Delta(x)) > \varepsilon + \inf_d E_x(d)\}$$

For some $\varepsilon > 0$, $\mathbb{P}(X \in A_\varepsilon) > 0$

On A_ε choose $\delta^*(x)$ so $E_x(\delta^*(x)) \leq E_x(\delta_\Delta(x)) - \varepsilon$

$$\delta^* = \delta_\Delta \text{ on } A_\varepsilon^c \Rightarrow E_x(\delta_\Delta(x)) - E_x(\delta^*(x)) \geq \varepsilon \mathbb{1}_{\{x \in A_\varepsilon\}}$$

Take expectations $\leadsto r_\Delta(\delta_\Delta) - r_\Delta(\delta^*) > \varepsilon \mathbb{P}(X \in A_\varepsilon)$ $\square$

Bayes Estimator

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Then $\delta_\Delta(x)$ is Bayes, with $r_\Delta(\delta_\Delta) < \infty$

iff $\delta_\Delta(x) \in \operatorname{argmin}_d \mathbb{E}[L(\theta, d) | X=x]$ a.e. x

Proof

$\mathbb{P}(\delta_\Delta(x) \notin \operatorname{argmin}) = 0$
wrt marginal

($\Leftarrow$) δ any other estimator

$$\mathbb{E}[L(\theta, \delta(x)) | X] \stackrel{\text{a.s.}}{\geq} \mathbb{E}[L(\theta, \delta_\Delta(x)) | X]$$

Marginalize $\leadsto r_\Delta(\delta) \geq r_\Delta(\delta_\Delta) < \infty$ (take $\delta = \delta_0$)

($\Rightarrow$) Assume $r_\Delta(\delta_\Delta) < \infty$ (o.w. δ_0 better)

$$\text{Define } \delta^*(x) = \begin{cases} \delta_\Delta(x) & \text{if } \delta_\Delta(x) \in \operatorname{argmin}_d (\dots) \\ \text{something better} & \text{if } \delta_\Delta(x) \notin \operatorname{argmin}_d (\dots) \end{cases}$$

$$\Rightarrow \mathbb{P}\left(\underbrace{\mathbb{E}[L(\theta, \delta^*(x)) | X]}_{\downarrow} < \underbrace{\mathbb{E}[L(\theta, \delta_\Delta(x)) | X]}_{\uparrow}\right) > 0$$
$$r_\Delta(\delta^*) = \mathbb{E}[\dots] < \mathbb{E}[\dots] = r_\Delta(\delta_\Delta)$$

Posterior Mean

If $L(\theta, d) = (g(\theta) - d)^2$ then the Bayes estimator is the posterior mean:

$$\begin{aligned}\mathbb{E}[(g(\theta) - d)^2 | X] \\&= \mathbb{E}[(g(\theta) - \mathbb{E}[g(\theta) | X] + \mathbb{E}[g(\theta) | X] - d)^2 | X] \\&= \text{Var}(g(\theta) | X) + (\mathbb{E}[g(\theta) | X] - d)^2\end{aligned}$$

(why is the cross-term 0?)

$$\Rightarrow \delta_{\Delta}(x) = \mathbb{E}[g(\theta) | X = x]$$

Weighted sq. error:

$$L(\theta, d) = w(\theta) (g(\theta) - d)^2$$

e.g. $(\frac{\theta-d}{\theta})^2$
sq. rel. error

$$\begin{aligned}\mathbb{E}[(d - g(\theta))^2 w(\theta) | X] \\&= d^2 \mathbb{E}[w(\theta) | X] - 2d \mathbb{E}[w(\theta)g(\theta) | X] \\&\quad + \mathbb{E}[w(\theta)g(\theta)^2 | X]\end{aligned}$$

no dep. on d

$$\text{min at } d = \frac{\mathbb{E}[w(\theta)g(\theta) | X]}{\mathbb{E}[w(\theta) | X]} \quad (= \delta_{\Delta}(x))$$

Example: Beta-Binomial

$$X | \theta \sim \text{Binom}(n, \theta) = \theta^x (1-\theta)^{n-x} \binom{n}{x}$$

$$\theta \sim \text{Beta}(\alpha, \beta) = \theta^{\alpha-1} (1-\theta)^{\beta-1} \underbrace{\frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha+\beta)}}_{\text{normalizing const.}}$$

θ is r.v. here

Marginal dist. of X called Beta-Binomial

Posterior:

$$\lambda(\theta | x) = \lambda(\theta) p_{\theta}(x) / q(x)$$

we can drop factors that don't depend on θ $\rightarrow$

$$\propto_{\theta} \theta^{\alpha-1} (1-\theta)^{\beta-1} \theta^x (1-\theta)^{n-x}$$
$$= \theta^{x+\alpha-1} (1-\theta)^{n-x+\beta-1}$$

$$\Rightarrow \theta | X=x \sim \text{Beta}(x+\alpha, n-x+\beta)$$

$$\mathbb{E}[\theta | x] = \frac{x+\alpha}{n+\alpha+\beta}$$

convex combo of $\frac{x}{n}$, $\frac{\alpha}{\alpha+\beta}$ $\rightarrow$

$$\overset{\text{UMVU}}{=} \frac{x}{n} \cdot \frac{n}{n+\alpha+\beta} + \underbrace{\frac{\alpha}{\alpha+\beta}}_{\text{Prior Expectation}} \cdot \underbrace{\frac{\alpha+\beta}{n+\alpha+\beta}}_{\left(1 - \frac{n}{n+\alpha+\beta}\right)}$$

Interp.: $k = \alpha + \beta$ "pseudo-trials," α successes

(Recall $\frac{x+3}{n+6}$ from Lec. 2)

Example: Normal mean

$$X|\theta \sim N(\theta, \sigma^2) \propto_{\theta} e^{-(x-\theta)^2/2\sigma^2}$$

$$\sim N(\mu, \tau^2) \propto_{\theta} e^{-(\theta-\mu)^2/2\tau^2}$$

$$\lambda(\theta|x) \propto_{\theta} \exp\left\{-\frac{(x-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu)^2}{2\tau^2}\right\}$$

$$\propto_{\theta} \exp\left\{\frac{x\theta}{\sigma^2} - \frac{\theta^2}{2\sigma^2} - \frac{\theta^2}{2\tau^2} + \frac{\theta\mu}{\tau^2}\right\}$$

$$= \exp\left\{\theta \underbrace{\left(\frac{x}{\sigma^2} + \frac{\mu}{\tau^2}\right)}_b - \theta^2 \underbrace{\left(\frac{\sigma^{-2} + \tau^{-2}}{2}\right)}_{a^2}\right\}$$

Complete square:

$$a^2\theta^2 - b\theta = \left(\theta a - \frac{b}{2a}\right)^2 - c(a, b)$$

$$= \left(\theta - \frac{b}{2a^2}\right)^2 \cdot a^2 - c(a, b)/a$$

$$\rightarrow \propto_{\theta} \exp\left\{-\left(\theta - \frac{x\sigma^{-2} + \mu\tau^{-2}}{\sigma^{-2} + \tau^{-2}}\right)^2 / 2(\sigma^{-2} + \tau^{-2})^{-1}\right\}$$

$$\propto_{\theta} N\left(\underbrace{\frac{x\sigma^{-2} + \mu\tau^{-2}}{\sigma^{-2} + \tau^{-2}}}_{\text{precision-weighted average of } x, \mu}, \frac{1}{\sigma^{-2} + \tau^{-2}}\right)$$

precision-weighted
average of x, μ

$$\mathbb{E}[\theta|x] = x \cdot \frac{\sigma^{-2}}{\sigma^{-2} + \tau^{-2}} + \mu \cdot \frac{\tau^{-2}}{\sigma^{-2} + \tau^{-2}}$$

Gaussian iid sample

$$\theta \sim N(\mu, \tau^2), \quad X_i | \theta \stackrel{iid}{\sim} N(\theta, \sigma^2), \quad i=1, \dots, n$$

$$\text{Suff. reduction} \rightsquigarrow \bar{X} | \theta \sim N(\theta, \frac{\sigma^2}{n})$$

$$\begin{aligned} \Rightarrow \mathbb{E}[\theta | X] &= \bar{X} \cdot \frac{n\sigma^{-2}}{n\sigma^{-2} + \tau^{-2}} + \mu \cdot \frac{\tau^{-2}}{n\sigma^{-2} + \tau^{-2}} \\ &= \bar{X} \cdot \frac{n}{n + \sigma^2/\tau^2} + \mu \cdot \frac{\sigma^2/\tau^2}{n + \sigma^2/\tau^2} \end{aligned}$$

Interp: $k = \sigma^2/\tau^2$ pseudo-observations, mean μ

If $n \gg k$, "data swamps prior"

If $n \ll k$, "prior swamps data"

[Note in both examples :

- Prior & Likelihood have similar fcn. form
- Posterior comes from same exp. f.m. as prior]

If the posterior is from the same family as the prior, we say the prior is conjugate to the likelihood.

Most common in exp. families

Conjugate Priors

Suppose $X_i | \eta \stackrel{\text{iid}}{\sim} p_\eta(x) = e^{\eta' T(x) - A(\eta)} h(x)$ $\eta \in \Xi \subseteq \mathbb{R}^s$
 $i=1, \dots, n$

For carrier $\lambda_0(\eta)$, define s+1-dim family:

$$\lambda_{\mu, k}(\eta) = e^{k\mu' \eta - kA(\eta) - B(k\mu, k)} \lambda_0(\eta)$$

Suff. stat $\begin{pmatrix} \eta \\ -A(\eta) \end{pmatrix} \in \mathbb{R}^{s+1}$ Nat. param. $\begin{pmatrix} k\mu \\ k \end{pmatrix}$

$$\begin{aligned} \Rightarrow \lambda(\eta | x_1, \dots, x_n) &\propto \eta \left(\prod_{i=1}^n e^{\eta' T(x_i) - A(\eta)} h(x_i) \right) \\ &\quad \cdot e^{k\mu' \eta - kA(\eta) - B(k\mu, k)} \lambda_0(\eta) \\ &= \alpha_\eta e^{(k\mu + \sum T(x_i))' \eta - (k+n)A(\eta)} \lambda_0(\eta) \\ &= \lambda_{\mu_{\text{post}}, k+n}(\eta) \end{aligned}$$

where

$$\mu_{\text{post}} = \frac{k\mu + n\bar{T}}{k+n}, \quad \bar{T}(X) = \frac{1}{n} \sum_{i=1}^n T(x_i)$$

often Bayes est. for $\mathbb{E}_\eta T$

$$\text{then } \mu_{\text{post}} = \underset{\substack{\uparrow \\ \text{UMVUE from} \\ \text{data}}}{\bar{T}} \cdot \frac{n}{k+n} + \mu \cdot \frac{k}{k+n}$$

$\uparrow$
 "UMVUE" from
 "pseudo data"

Conjugate Prior Examples

Likelihood	Prior
$X_i \theta \sim \text{Binom}(n, \theta)$ $= \theta^x (1-\theta)^{n-x} \binom{n}{x}$	$\theta \sim \text{Beta}(\alpha, \beta)$ $= \theta^{\alpha-1} (1-\theta)^{\beta-1} \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha+\beta)}$
$X_i \theta \sim N(\theta, \sigma^2)$ (σ^2 known) $= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\theta-x)^2}{2\sigma^2}}$	$\theta \sim N(\mu, \tau^2)$ $= \frac{1}{\sqrt{2\pi}\tau} e^{-\frac{(\theta-\mu)^2}{2\tau^2}}$
$X_i \theta \sim \text{Pois}(\theta)$ $x=0,1,\dots$ $= \frac{\theta^x e^{-\theta}}{x!}$	$\theta \sim \text{Gamma}(v, s)$ $\theta > 0$ $= \frac{1}{\Gamma(v) s^v} \theta^{v-1} e^{-\theta/s}$

Gamma / Poisson :

$$\lambda(\theta | x) \propto_{\theta} \theta^{v-1 + \sum x_i} e^{-(s' + n)\theta}$$

$$= \text{Gamma}(v + \sum x_i, (s' + n)^{-1})$$

$$\Rightarrow k = s^{-1}, \quad \mu = vs$$

$$\lambda_0(\theta) = \theta^{-1} \quad (\text{not normalizable})$$

Flexibility of Bayes

Any $\Delta, \mathcal{P}, L, g(\theta): \mathcal{J}_\Delta$ defined straightforwardly

$$\mathcal{J}_\Delta(x) = \operatorname{argmin}_d \int L(\theta, d) \lambda(\theta|x) d\theta$$

Problem reduced to (possibly hard) computation

Posterior is "one stop shop" for all answers

No need for:

- special family structure (exp. fam. / complete s.s.)
- special estimator (U-estimable)
- convex or nice L

$\Rightarrow$ Highly expressive modeling & estimation

Caveat: Limited by ability to do computations